

CS-23334 FUNDAMENTALS OF DATA SCIENCE

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Experiment 9

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9. Experiment to understand KNN algorithm for a given dataset

Aim:

To conduct experiment to understand KNN algorithm for a given dataset

Description:

Understand the KNN algorithm for the dataset given.

Algorithm:

Step 1: Select Features and Preprocess the Data

Step 2: Normalize data–

Step 3: Apply the KNN Algorithm and Fit the Model

Step 4: Visualize Clusters and Centroids

Step 5: Interpret Cluster Assignments and Evaluate Results

About Dataset:

This dataset contains customer demographic and behavioral data, including Customer ID, Gender, Age, Annual Income (in thousands), and a Spending Score from 1 to 100.

Code With Output:

```
import numpy as np
import pandas as pd

df=pd.read_csv(r'D:\REC 2nd Year\Data Science\Data Sets\Iris KNN.csv')
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 5 columns):
#   Column          Non-Null Count  Dtype  
---  -
0   sepal.length    150 non-null   float64
1   sepal.width     150 non-null   float64
2   petal.length    150 non-null   float64
3   petal.width     150 non-null   float64
4   variety         150 non-null   object  
dtypes: float64(4), object(1)
memory usage: 6.0+ KB
```

```
df.variety.value_counts()
```

```
variety
Setosa      50
Versicolor  50
Virginica   50
Name: count, dtype: int64
```

```
df.head()
```

	sepal.length	sepal.width	petal.length	petal.width	variety
0	5.1	3.5	1.4	0.2	Setosa
1	4.9	3.0	1.4	0.2	Setosa
2	4.7	3.2	1.3	0.2	Setosa
3	4.6	3.1	1.5	0.2	Setosa
4	5.0	3.6	1.4	0.2	Setosa

```
features=df.iloc[:, :-1].values
label=df.iloc[:, 4].values
```

```

from sklearn.model_selection import train_test_split
from sklearn.neighbors import KNeighborsClassifier

xtrain,xtest,ytrain,ytest=train_test_split(features,label,test_size=0.
2, random_state=26)
model_KNN=KNeighborsClassifier(n_neighbors=5)
model_KNN.fit(xtrain,ytrain)

KNeighborsClassifier()

print(model_KNN.score(xtrain,ytrain))
print(model_KNN.score(xtest,ytest))

```

```

0.9666666666666667
0.9666666666666667

```

```

from sklearn.metrics import confusion_matrix
confusion_matrix(label,model_KNN.predict(features))

```

```

array([[50,  0,  0],
       [ 0, 47,  3],
       [ 0,  2, 48]])

```

```

from sklearn.metrics import classification_report
print(classification_report(label,model_KNN.predict(features)))

```

	precision	recall	f1-score	support
Setosa	1.00	1.00	1.00	50
Versicolor	0.96	0.94	0.95	50
Virginica	0.94	0.96	0.95	50
accuracy			0.97	150
macro avg	0.97	0.97	0.97	150
weighted avg	0.97	0.97	0.97	150

Result:

Thus python program to understand KNN algorithm for dataset is conducted successfully